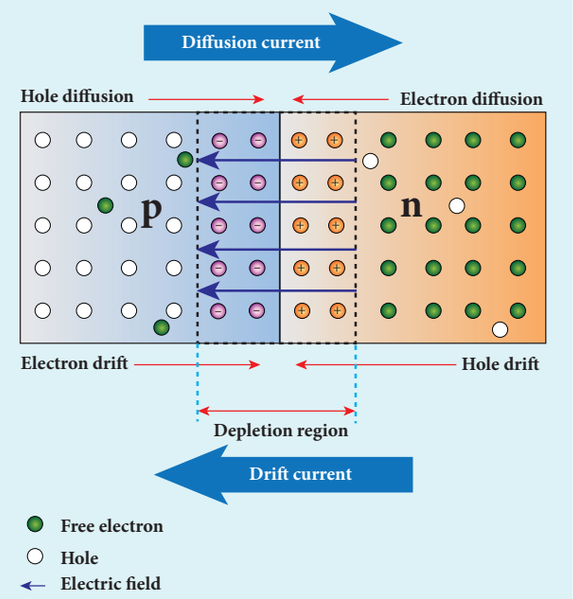


Figure 10.9 Formation of *p*–*n* junction

the negatively charged layer in the *p*-side in the depletion region as shown in the Figure 10.9. This electric field makes electrons in the *p*-side drift into the *n*-side and the holes in the *n*-side into the *p*-side. **The electric current produced due to the motion of the minority charge carriers by the electric field is known as drift current**. The diffusion current and drift current flow in opposite directions.

Though drift current is less than diffusion current initially, equilibrium is reached between them at a particular time. With each electron (or hole) diffusing across the junction, the strength of the electric field increases thereby increasing the drift current till the two currents become equal. Hence at equilibrium, there is no net electric current across the junction. Thus, a *p*-*n* junction is formed.

ii) Junction potential or barrier potential

The movement of charge carriers across the junction takes place only to a certain point beyond which the depletion layer acts like a barrier to further diffusion of free charges across the junction. This is due to the fact that the immobile ions on both sides establish an electric potential difference across the junction. Therefore, an electron trying to diffuse into the interior of the depletion region encounters a wall of negative ions repelling it backwards. If the free electron has enough energy, it can break through the wall and enter into the *p*-region, where it can recombine with a hole and create another negative ion.

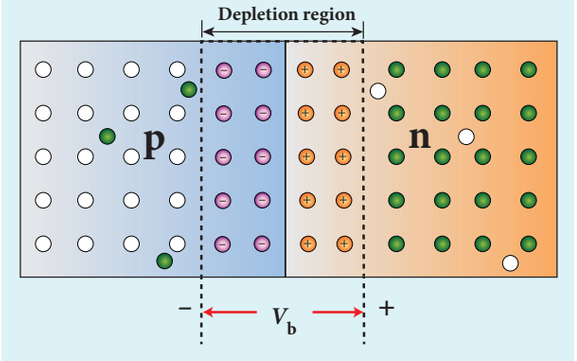


Figure 10.10 Barrier potential formed across the junction

The strength of the electric potential difference across the depletion region keeps on increasing with the crossing of each electron until equilibrium is reached; at this point, the internal repulsion of the depletion layer stops further diffusion of free electrons across the junction. **This difference in potential across the depletion layer is called the barrier potential (V_b)** as shown in Figure 10.10. At 25 ^{0}C, this barrier potential is approximately 0.7 V for silicon and 0.3 V for germanium.

10.3.2 P-N Junction diode

A *p*-*n* junction diode is formed when a *p*-type semiconductor is fused with an *n*-type semiconductor. It is a device with single *p*-*n* junction as shown in Figure 10.11(a) and its circuit symbol is shown in Figure 10.11(b).

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